

Chenyu Fang, Ph.D.

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SUMMARY

AI/ML Engineer (Ph.D.) with 3+ years of applied ML/research experience building LLM applications end to end: document AI pipelines, RAG, tool-using agents with guardrails, and the evaluation that keeps them honest. Strongest in LLM evaluation, verifying model output, and backend integration.

TECHNICAL SKILLS

LLM systems: RAG, hybrid retrieval, AI agents, LangGraph, MCP, LLM evaluation, guardrails, vision-language models, LoRA fine-tuning (Hugging Face, PEFT), vLLM

Backend and deployment: FastAPI, PostgreSQL/pgvector, job queues, Docker, Kubernetes (GKE, kind), Azure (Container Apps), AWS, GCP (Vertex AI, Cloud Run), ONNX, GitHub Actions, pytest

ML and languages: Python, PyTorch, scikit-learn, XGBoost, LightGBM, SQL, C++, Linux, Git

SELECTED PROJECTS

AI Tender Evaluation Assistant

02/2026 - Present

Team project, Techlent Machine Learning Engineer Fellowship (training program)

Demo: tender-eval-ai.github.io/tender-evaluation-assistant | Code: github.com/tender-eval-ai/tender-evaluation-assistant

- Co-built an LLM bid-review system with a second engineer; I owned the FastAPI backend, LLM gateway and bid checks. Models only read, a deterministic rule engine decides, and every value is cited to its page.
- Built Postgres-backed jobs with checkpoints, approval pauses and rate-limit backoff; chose Procrastinate over LangGraph after a 12-scenario fault-injection harness showed about 5 s recovery, **0 lost jobs** and no repeated model calls.
- Verified every extracted value by text-layer match or an independent second read, sending failures to human review, and added a bounded search agent tested against prompt injection.
- On a synthetic tender, rule drafting matched 15/15 checklist items with no unverified rules, at \$0.02.
- Deployed on Azure Container Apps with Entra ID sign-in and OIDC deploys that store no secrets.

MLE/AIE Interview Coach

Demo: coach.cyfang.org | Code: github.com/chenyufang-data/mle-aie-interview-coach

- Built and deployed a voice mock-interview app grounded in rubrics and tailored to each user's resume and target job.
- Fine-tuned Qwen3-4B with LoRA on 598 Claude-labeled synthetic answers (~20% held out per seed); its grades agreed with Claude's at **0.94 on a 0-1 scale (QWK)** across five seeds, and vLLM served a grading call in 30 ms (p95). For hosted grading, chose DeepSeek at ~1/200th Claude's cost, with 94% of grades within one point of Claude.
- Raised how often the right passage ranks in the top 5 (Recall@5) from 72% to 84% with hybrid BM25 + BGE search.

Gridcast | NYISO Day-Ahead Load Forecasting

Demo: gridcast.cyfang.org | Code: github.com/chenyufang-data/gridcast

- Built a live day-ahead load forecasting service for 11 NYISO zones: a PyTorch Temporal Fusion Transformer served as ONNX through FastAPI on GCP, with daily ingestion, self-scoring and a LightGBM fallback.
- On par with NYISO's own forecast: **4.80% vs. 4.85% average hourly error** (MAPE; difference not significant).
- Deployed it on Kubernetes (kind in CI on every push, a 72-hour GKE Autopilot trial) with a digest-pinned model image.

EXPERIENCE

University of Oklahoma

Norman, OK

Research Assistant (part-time)

02/2026 - Present

Rare-Signal Benchmark: XGBoost vs. LorentzNet vs. Particle Transformer

Code: github.com/chenyufang-data/bsm-higgs-ml-benchmark

- Benchmarked XGBoost against a re-implemented LorentzNet graph network and a Particle Transformer on identical splits over three seeds; both networks beat XGBoost's AUC under paired bootstrap intervals.
- LorentzNet **matched the transformer at a tenth of its size** with 2x faster inference; XGBoost trailed both by under 0.01 AUC at about 1/100th the training cost.

PhD Researcher

08/2018 - 12/2025

Code: github.com/chenyufang-data/hep-ml-pipeline

- Developed Python/C++ pipelines and XGBoost classifiers for multi-million-event simulated datasets; used SHAP and feature ablations to evaluate signal separation, recovering **41.2% of signal at a 7.7% false-positive rate**.

Research outputs: Two first-author physics preprints (2025): [arXiv:2511.08526](https://arxiv.org/abs/2511.08526), [arXiv:2511.19604](https://arxiv.org/abs/2511.19604).

EDUCATION

University of Oklahoma - PhD in Physics

Norman, OK | 08/2018 - 12/2025

Huazhong University of Science and Technology - B.S. in Physics

Wuhan, China | 09/2014 - 06/2018